Take-Home Assignment: Grokking in Neural Networks

Time limit: 3 hours | **Deliverable:** Jupyter Notebook or Python Package

THE PROBLEM

Grokking is a surprising phenomenon where a neural network first memorizes a small, algorithmic dataset (overfitting) to achieve perfect training accuracy; then, after much longer training, suddenly shifts to learning the underlying general rule, leading to a sharp jump in test accuracy.

It represents a transition from *lazy memorization* to *deeper generalization*, challenging typical stopping criteria and showing that models can find elegant solutions long after appearing to be "done."

Your task: Reproduce this phenomenon yourself and study it in as much detail as possible.

APPROACH

Design and run an experiment where a model trained on a small, algorithmic dataset exhibits grokking-like behavior. You may, for example:

- **Generate a synthetic dataset** with a simple underlying rule (e.g., modular arithmetic or combinatorial tasks).
- **Train a small MLP** (or another suitable architecture) for many more steps than needed to reach perfect training accuracy.
- **Log metrics across a dense set of checkpoints**, so that any delayed jump in generalization can be clearly visualized.
- **Inspect internal dynamics** (e.g., weight norms, activations, or gradients) to correlate internal changes with late improvement in test performance.

You are free to choose the dataset, architecture, optimizer, and visualization strategy. Focus on the research aspects you find most compelling.

DELIVERABLES

1. Code

A script or notebook that constructs the dataset, trains the model for an extended duration, and logs relevant metrics and diagnostics.

2. Results

Plots of training and test loss/accuracy over the full training run, plus additional summaries that illuminate internal model changes (e.g., weight norms).

3. Brief Commentary

A short write-up (or Markdown section) describing:

- Your experimental setup and design choices.
- Evidence of grokking-like behavior and how you diagnosed it.
- Your hypothesis on the dynamics based on the evidence.

EXTENSIONS

If time permits, consider exploring:

- **Variations:** Different algorithmic tasks or dataset sizes.
- **Optimization:** Changing the optimizer or regularization (e.g., Weight Decay).
- **Probing:** Looking at feature correlations or layer-wise behavior.

Success means that **you learn something** and **you teach us something** about grokking through your experiment. Good luck!